Cartesian Equation of a Line in R^2

Recall:

Vector Equation: $\vec{r} = \vec{r}_0 + t\vec{m}$, $t \in \Re$

Vector Equation in Component Form: $(x, y) = (x_0, y_0) + t(a, b)$, $t \in \Re$

Parametric Equations: $\begin{matrix} x = x_0 + ta \\ y = y_0 + tb \end{matrix}$, $t \in \Re$

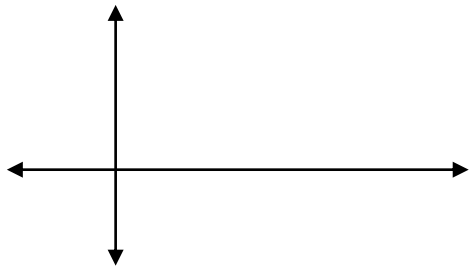
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Determine an equation of a line in **standard form** that contains the point (0 , 5) and has direction vector $\vec{m} = (3,4)$.

A _____ is a vector that is _____ to a given vector.

E.g. Determine a normal vector to the vector $\vec{m} = (3,4)$.

Example 1: Determine the Cartesian equation of a line whose normal vector is $\vec{n} = (4,-2)$ and the line contains the point (4 , 5).



Derive using general form: Given point $P_0(x_0, y_0)$ and normal vector $\vec{n} = (A, B)$

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$$Ax + By + C = 0$$

Example 2: Determine the Cartesian equation of a line through the point (2 , -9) with the normal $\vec{n} = (-4,5)$.

Example 3: Given $\vec{m} = (3,2)$ and a point $P_0(4,-1)$,

a) Determine a vector equation of this line.

b) Determine the parametric equations of this line.

c) Determine the Cartesian equation of this line.

Symmetric Equation of a line in R^2

Using the parametric equations, we could create _____ and use that to create the Cartesian equation.